

# Paper Replication Results Summary

**Model:** cemoid quantum classifier, **L = 3 layers**, **P = 2 cemoid-block repetitions** **Task:** tic-tac-toe board classification (3-class) **Base circuit:** 9 qubits, 6 cemoid blocks, **54 trainable gate parameters** **Training:** Adam (lr = 0.03), 100 epochs × 30 steps, batch size 15; fixed data split (**DATA\_SEED = 2027**)

This document consolidates the two robustness experiments performed on the best L/P configuration identified by the evolutionary sweep:

- 1. **50-seed robustness** — how stable is the solution across random initialisations?
- 2. **Random gate insertion** — is the trained solution a unique global optimum, or one of many degenerate attractors?

It is preceded by the **L/P architecture sweep** that motivated the choice of configuration.

## Experiment 0 — L/P Architecture Sweep (1–7 × 1–7), agrees with [paper](#)

### Design

Before fixing a configuration, every combination of **L ∈ {1...7} layers** and **P ∈ {1...7} cemoid-block repetitions** was trained once under identical conditions (49 models). Each model records its full 100-epoch test-accuracy trajectory; the final-epoch accuracy is reported.

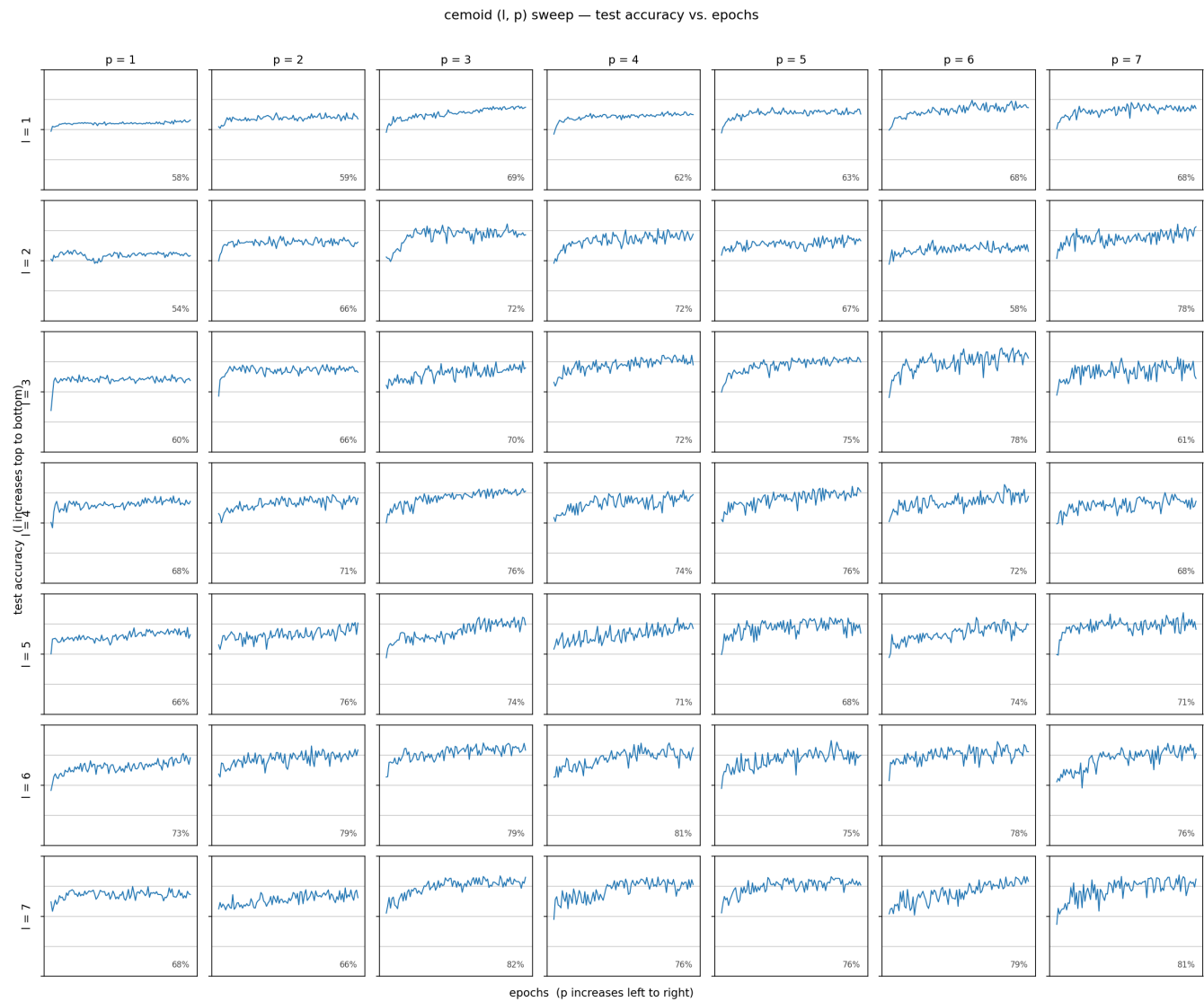
Final test accuracy grid (rows = L, cols = P)

| L \ P                   | P1    | P2                                   | P3    | P4    | P5    | P6    | P7    |
|-------------------------|-------|--------------------------------------|-------|-------|-------|-------|-------|
| L1                      | 0.580 | 0.590                                | 0.687 | 0.625 | 0.630 | 0.683 | 0.677 |
| L2                      | 0.543 | 0.655                                | 0.717 | 0.722 | 0.667 | 0.578 | 0.783 |
| L3                      | 0.595 | 0.663                                | 0.700 | 0.725 | 0.748 | 0.782 | 0.612 |
| L4                      | 0.680 | 0.707                                | 0.765 | 0.737 | 0.762 | 0.722 | 0.682 |
| L5                      | 0.662 | 0.758                                | 0.743 | 0.713 | 0.675 | 0.745 | 0.707 |
| L6                      | 0.728 | 0.793                                | 0.792 | 0.812 | 0.748 | 0.778 | 0.762 |
| L7                      | 0.682 | 0.655                                | 0.825 | 0.765 | 0.760 | 0.792 | 0.810 |
| Metric                  |       | Value                                |       |       |       |       |       |
| Models trained          |       | 49 (7 × 7)                           |       |       |       |       |       |
| Min / Max               |       | 0.543 (L2,P1) / <b>0.825 (L7,P3)</b> |       |       |       |       |       |
| Selected config (L3,P2) |       | 0.663                                |       |       |       |       |       |

Accuracy broadly increases with circuit depth (larger L and P), with the strongest single runs in the L6–L7 band (up to 0.825). **L=3, P=2** (0.663) was selected for the downstream robustness experiments as a compact, mid-

range configuration — large enough to classify well above chance, small enough (54 parameters) to train cheaply across 50 seeds and 70 gate-insertion runs.

Plot



Final test accuracy across the full  $L = 1-7 \times P = 1-7$  grid.

Experiment 1 — 50-Seed Robustness

Design

The fixed ( $L=3, P=2$ ) circuit is trained **50 times**, each with a different RNG seed for the initial gate parameters, while the data split is held constant. Each run records the full 100-epoch test-accuracy trajectory; the final epoch accuracy is the reported statistic.

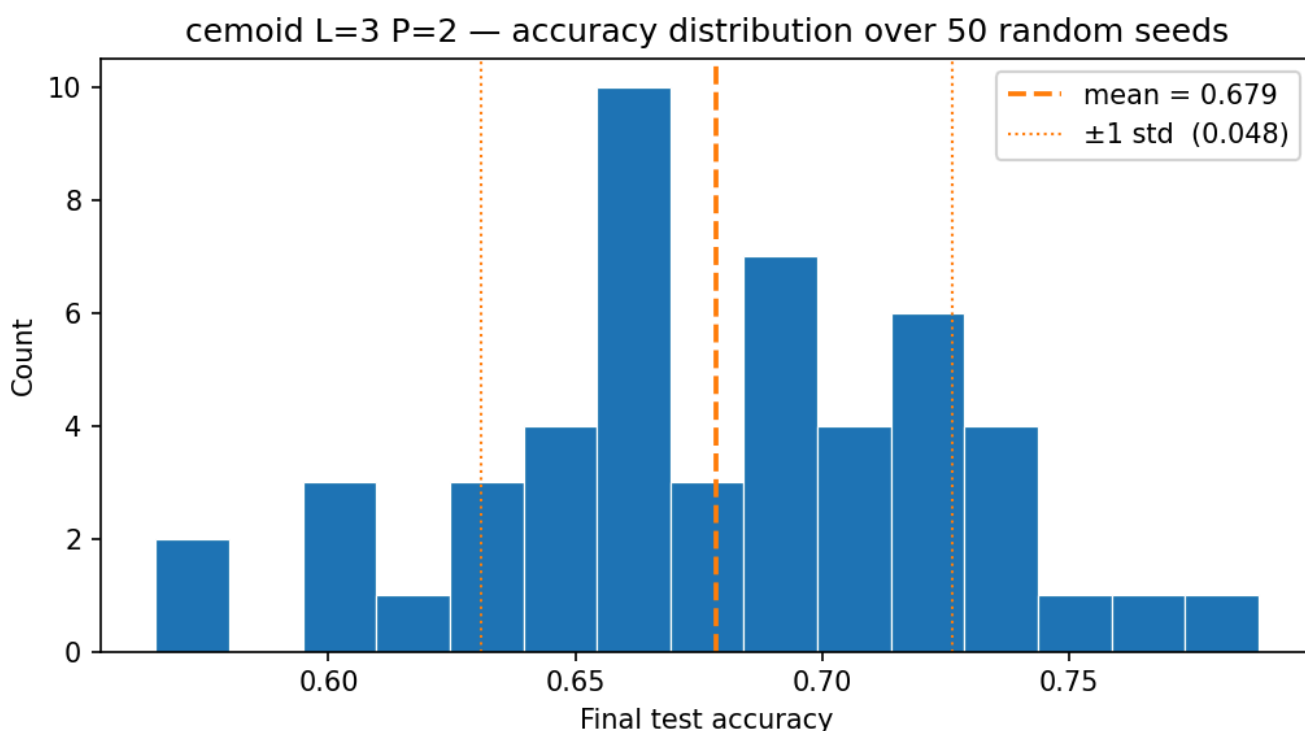
Statistics ( $n = 50$ )

| Metric                   | Value |
|--------------------------|-------|
| Mean final test accuracy | 0.679 |

| Metric               | Value                             |
|----------------------|-----------------------------------|
| Std (sample)         | 0.048                             |
| Median               | 0.680                             |
| Min / Max            | 0.565 / 0.788                     |
| Range                | 0.223                             |
| Best / worst seed    | seed 21 (0.788) / seed 17 (0.565) |
| Fraction $\geq 0.65$ | 76%                               |
| Fraction $\geq 0.70$ | 34%                               |

Random-guess baseline for a 3-class task is  $\approx 0.33$ , so all 50 runs are far above chance. The  $\sim 0.048$  spread shows the optimiser reliably reaches the 0.66–0.70 band but lands on different solutions of varying quality depending on initialisation.

## Plot



*Distribution of final test accuracy across the 50 random seeds for the fixed L=3, P=2 cemoid circuit.*

## Experiment 2 — Random Gate Insertion

### Hypothesis

If the trained L=3, P=2 solution is a strict global optimum, any extra rotation gate inserted anywhere — initialised at angle **0** (a perfect no-op) — should stay at 0 under gradient descent.

If the learned angles drift **away from zero**, the loss landscape contains other equally good attractors and the solution is not unique.

## Design

- **Slot skeleton:** 10 insertion slots  $\times$  9 qubits  $\times$  3 rotation types (RX/RY/RZ) = **270 candidate positions**.
- For each gate count  $N \in \{1, 2, 3, 5, 8, 13, 21\}$ ,  $N$  positions are drawn at random with a **fixed config seed = 42**, so the inserted topology is identical across all training seeds.
- Extra-gate angles always start at exactly 0; cemoid parameters use **10 independent training seeds** (0–9) per gate count.
- All parameters (cemoid + extra) are optimised jointly. **70 runs total** (7 gate counts  $\times$  10 seeds).

Statistics (10 runs per gate count)

| Extra gates N      | Mean acc     | Std          | Min   | Max   | Mean $ \theta_{\text{extra}} $ (rad) | Std $ \theta $ |
|--------------------|--------------|--------------|-------|-------|--------------------------------------|----------------|
| 1                  | 0.679        | 0.041        | 0.620 | 0.763 | 0.349                                | 0.320          |
| 2                  | 0.677        | 0.042        | 0.622 | 0.750 | 0.212                                | 0.165          |
| 3                  | 0.682        | 0.051        | 0.602 | 0.767 | 0.095                                | 0.064          |
| 5                  | 0.662        | 0.027        | 0.625 | 0.715 | 0.092                                | 0.033          |
| 8                  | 0.676        | 0.035        | 0.620 | 0.727 | 0.232                                | 0.166          |
| 13                 | 0.662        | 0.035        | 0.582 | 0.708 | 0.262                                | 0.164          |
| 21                 | 0.658        | 0.029        | 0.613 | 0.695 | 0.188                                | 0.053          |
| <b>pooled (70)</b> | <b>0.671</b> | <b>0.037</b> | —     | —     | —                                    | —              |

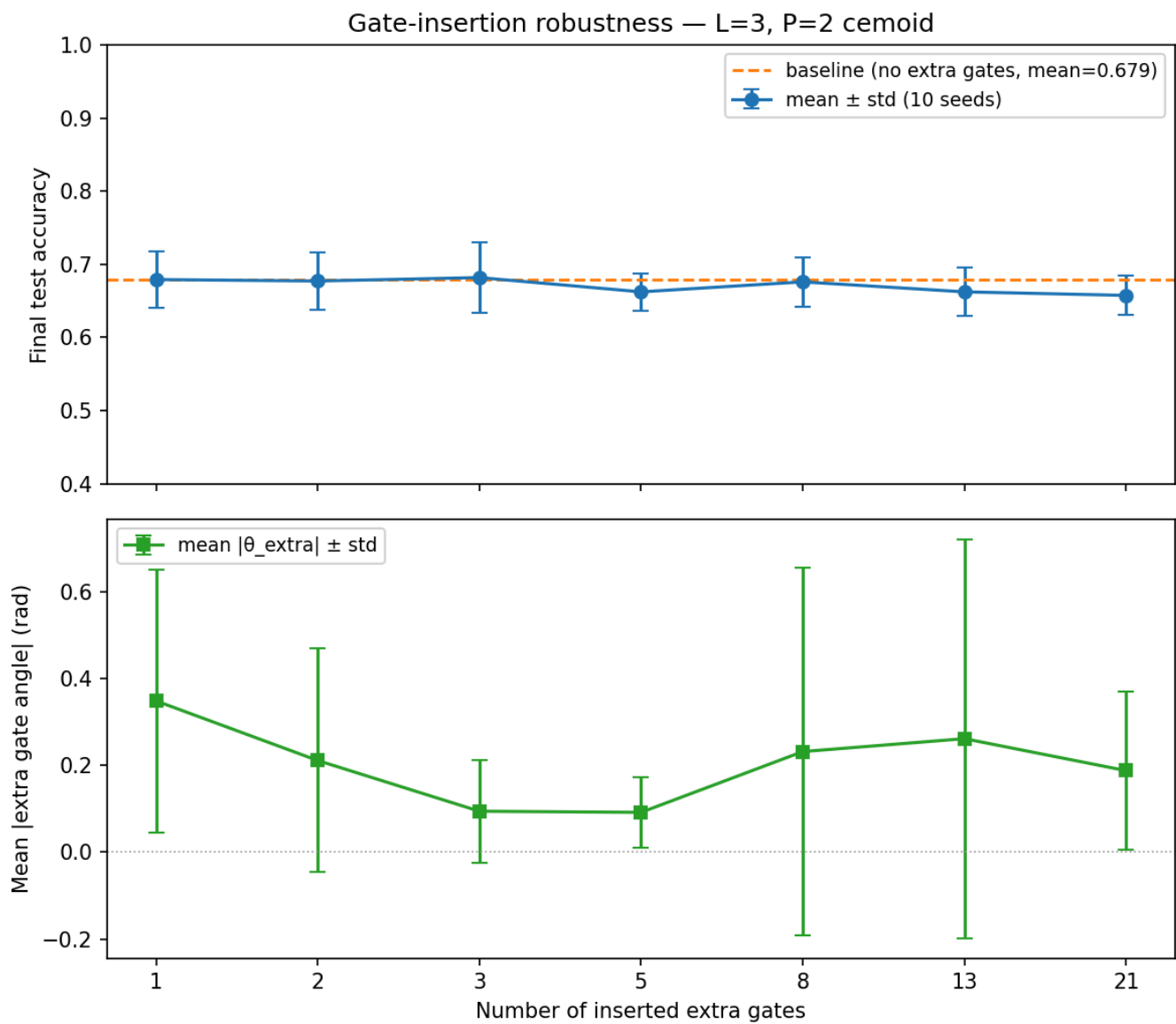
**Baseline (50 seeds, no extra gates): mean 0.679, std 0.048.**

Significance vs. baseline (Welch t-test)

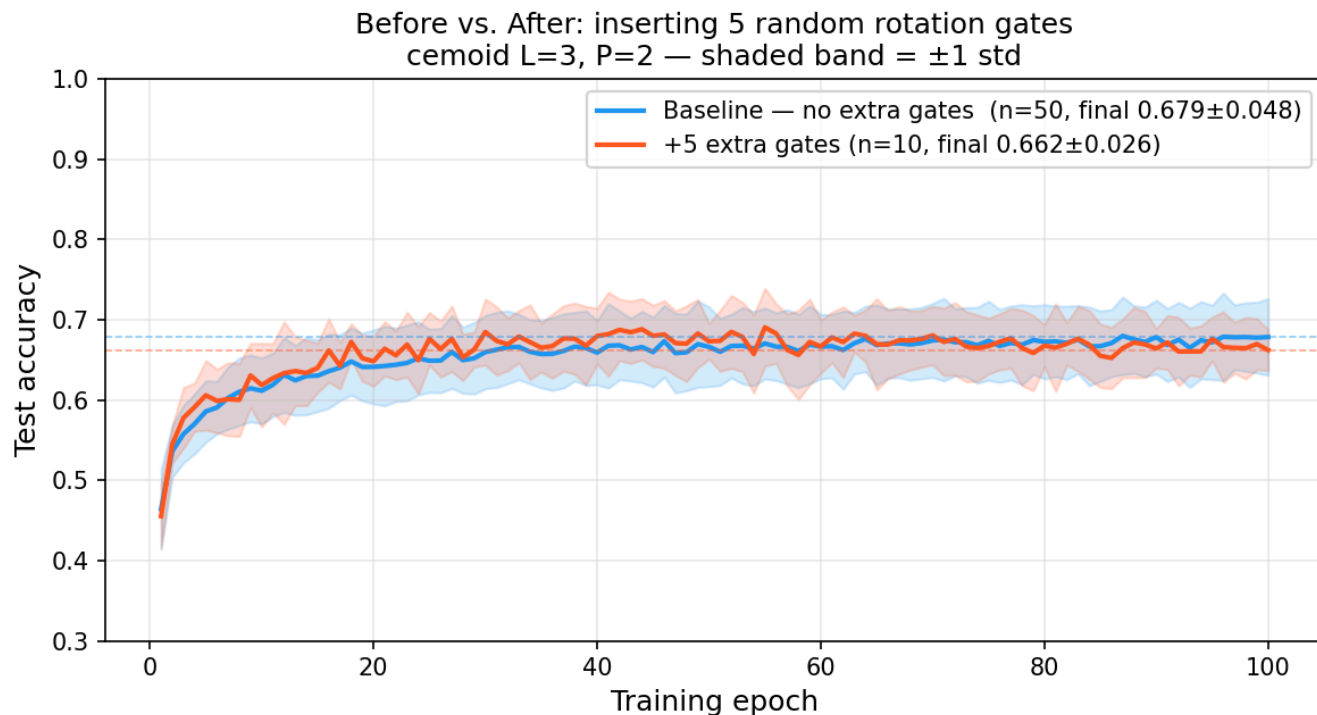
| N      | $\Delta\text{acc vs baseline}$ | t     | Significant? |
|--------|--------------------------------|-------|--------------|
| 1      | +0.001                         | −0.04 | no           |
| 2      | −0.001                         | +0.09 | no           |
| 3      | +0.003                         | −0.19 | no           |
| 5      | −0.016                         | +1.48 | no           |
| 8      | −0.003                         | +0.20 | no           |
| 13     | −0.016                         | +1.26 | no           |
| 21     | −0.021                         | +1.86 | no           |
| pooled | −0.008                         | +0.94 | no           |

All  $|t| < 2 \rightarrow$  **no statistically significant accuracy change** from adding up to 21 free parameters.

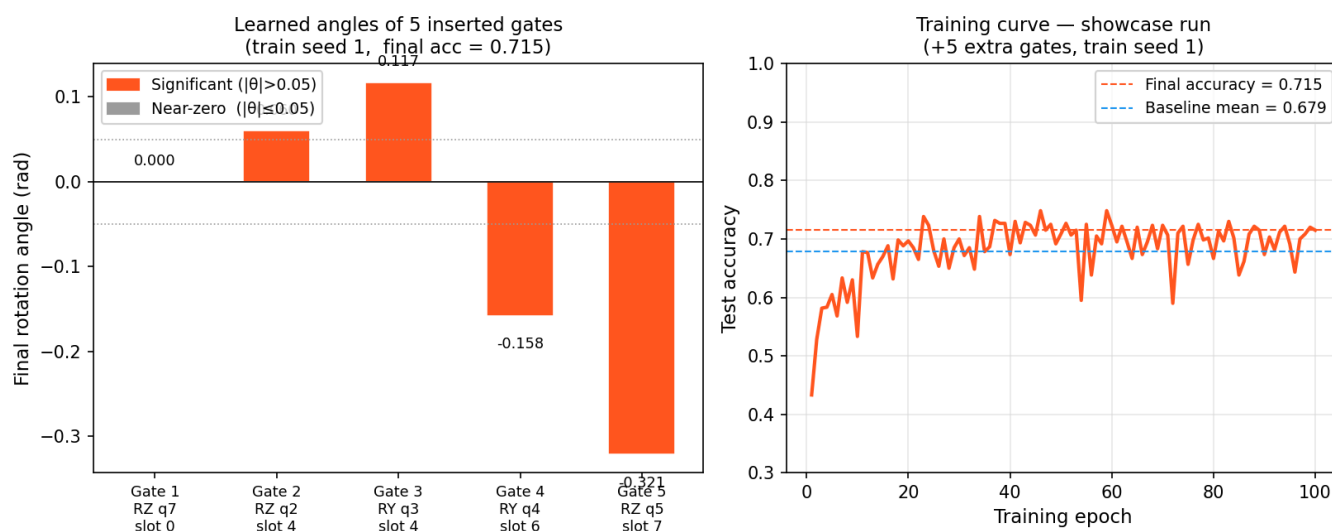
Plots



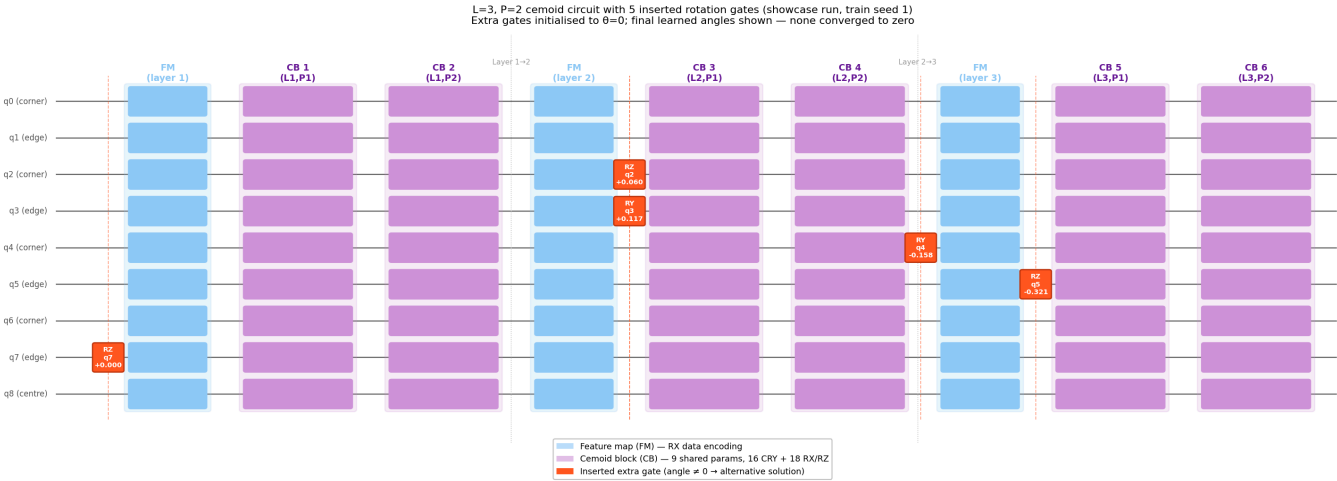
Top: mean test accuracy vs. number of inserted gates (flat, near baseline). Bottom: mean absolute extra-gate angle (consistently 0.09–0.35 rad, never zero).



Full 100-epoch trajectories: baseline (n=50) vs +5 extra gates (n=10) converge along nearly identical paths.



A single high-performing +5-gate run (train seed 1, final acc 0.715). Four of the five inserted gates learn non-trivial angles ( $|\theta| > 0.05$  rad).



*L=3, P=2 cemoid circuit with the 5 inserted gates highlighted; learned angles annotated.*

Key Findings

| Question                                  | Finding                                                                                                                           |
|-------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Is the solution stable across seeds?      | Yes — $0.679 \pm 0.048$ , all 50 runs well above chance (0.33).                                                                   |
| Does accuracy hold with extra gates?      | Yes — 0.658–0.682 across 1–21 gates; no significant change (all $ t  < 2$ ).                                                      |
| Do inserted gates converge to zero?       | <b>No</b> — mean $ \theta  = 0.09\text{--}0.35$ rad, well above zero.                                                             |
| Is the trained solution uniquely optimal? | <b>No</b> — gates initialised at 0 drift to non-zero values, indicating degenerate attractors / flat directions in the landscape. |
| Practical robustness for NISQ hardware?   | Yes — the circuit absorbs extra free parameters without performance loss.                                                         |

**Interpretation.** The two experiments are complementary. The seed sweep shows the optimiser reaches a consistent accuracy band from many starting points; the gate-insertion sweep shows those endpoints are *not* a single global minimum but a family of equivalent solutions connected by flat directions. The model is robust and expressive, but the specific gate angles of any single trained instance should not be over-interpreted as physically meaningful.

Reproducibility

| Item                    | Path                                                              |
|-------------------------|-------------------------------------------------------------------|
| L/P sweep raw histories | <code>histories/history_l{1-7}_p{1-7}.json</code>                 |
| L/P sweep script        | <code>sweep.py</code>                                             |
| 50-seed raw histories   | <code>robustness_histories/seed_000.json ... seed_049.json</code> |

| Item                                                    | Path                                                            |
|---------------------------------------------------------|-----------------------------------------------------------------|
| Gate-insertion raw results                              | <a href="#">gate_insertion_results/ng{N}_cs042_ts{0-9}.json</a> |
| Seed-sweep script                                       | <a href="#">seed_robustness.py</a>                              |
| Gate-insertion script                                   | <a href="#">gate_insertion.py</a>                               |
| Figure-generation script                                | <a href="#">make_report_figures.py</a>                          |
| Detailed write-up (incl. pending EA-circuit comparison) | <a href="#">gate_insertion_report.md</a>                        |

*Statistics in this summary were recomputed directly from the raw JSON result files. The EA-evolved-circuit comparison (Section 6 of [gate\\_insertion\\_report.md](#)) is still pending cluster completion — [ea\\_robustness\\_histories/](#) is currently empty.*